

MANNAT PAL

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EDUCATION

Vellore Institute of Technology

Integrated M.Tech in Computer Science and Engineering
GPA: 9.0

Oct 2022 – Present

MASD Public School

Class XII
Percentage: 86.4%

Panipat, Haryana
Mar 2020 – Jun 2021

Air Force School

Class X
Percentage: 89.6%

Gorakhpur, Uttar Pradesh
Mar 2018 – Jun 2019

EXPERIENCE

Cybersecurity Analyst Intern (Completed)

Sequaretek IT Solutions Pvt. Ltd.

Nov 2025 - Dec 2025
Mumbai, India

- Orchestrated a **behaviour-based malware analysis** pipeline leveraging honeypot telemetry and large language models, attaining a **74% detection accuracy**, crucial for rapid threat identification within the network.
- Integrated automated pre-analysis modules, pulling **150+ behavioural indicators** and file hashes per sample, speeding up malware classification **workflows by 15%** and helping the members' efficiency.
- Generated structured malware analysis reports and IoCs using **GPT-OSS: latest and Mistral:7b**, and validated threat intelligence via **VirusTotal**, analysing 500+ malware samples.

PROJECTS

Android Malware Detection Using Machine Learning

(Github)

- Constructed a machine learning pipeline for Android malware detection; **achieved 92.86% accuracy** in identifying malicious applications, streamlining threat triage and accelerating security risk assessments for critical vulnerabilities.
- Performed static analysis on **429 Android APKs** by extracting manifest-level features, including permissions, intents, and component metadata, using **Python and Androguard**.
- Designed an **imbalance-aware malware triage pipeline** to reduce false positives and prioritise high-risk applications for analyst-driven investigation.

ThreatCopilot

(Github)

- Developed an **AI-powered SOC investigation assistant using FastAPI and Elasticsearch**, enabling analysts to query security logs through natural language and perform flexible filtering, aggregation, and attacker analysis.
- Designed an **NLP-based intent parsing layer** (rule-based + optional LLM) to convert analyst queries into structured Elasticsearch searches with multi-turn session context.
- Integrated **MITRE ATT&CK mapping** (T1110 – Brute Force) with threshold-based detection to identify suspicious login activity and high-risk source IPs, **simulating real-world SOC investigations**.

SKILLS

Security Tools: Wireshark, Nmap, Burp Suite, Metasploit, OWASP ZAP, SQLMap, Nikto, Androguard, APKTool, VirusTotal, Malware Static Analysis, Splunk, Snort

Machine Learning: Scikit-learn (Random Forest, XGBoost, Naive Bayes), Pandas, NumPy, Matplotlib

Programming, Cloud & Dev Tools: C++, Python, HTML, CSS, SQL, Docker, AWS, Elasticsearch, Git, GitHub

Core Security Concepts: Threat Intelligence, Incident Response, Malware Analysis, Vulnerability Assessment, MITRE ATT&CK Framework

Computer Science Fundamentals: Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

EXTRA-CURRICULAR ACTIVITIES

- Class Representative:** Served as primary liaison between faculty and 60+ students, coordinating academic communications, resolving student concerns, and supporting smooth course administration.
- Event Management Lead, Mozilla Firefox Club:** Organized 5+ technical events with 100+ participants, coordinating cross-team logistics.
- Content and Editorial Lead, Cyber Warriors Club:** Drove cybersecurity awareness initiatives by curating technical content and coordinating knowledge-sharing sessions.